Security Analyst

ASSIGNMENT REPORT

Networking Fundamentals

Full name	*Kagia Joseph Njuguna*
Admission number	*CS-SA13-26022*
Email	*kagiajoseph72@gmail.com*
Week and assignment	*Week 4: Assignment 2: HTB - Networking Fundamentals*
Platform	*Hack The Box,*
Date submitted	*11 October 2026*

1. Module completion

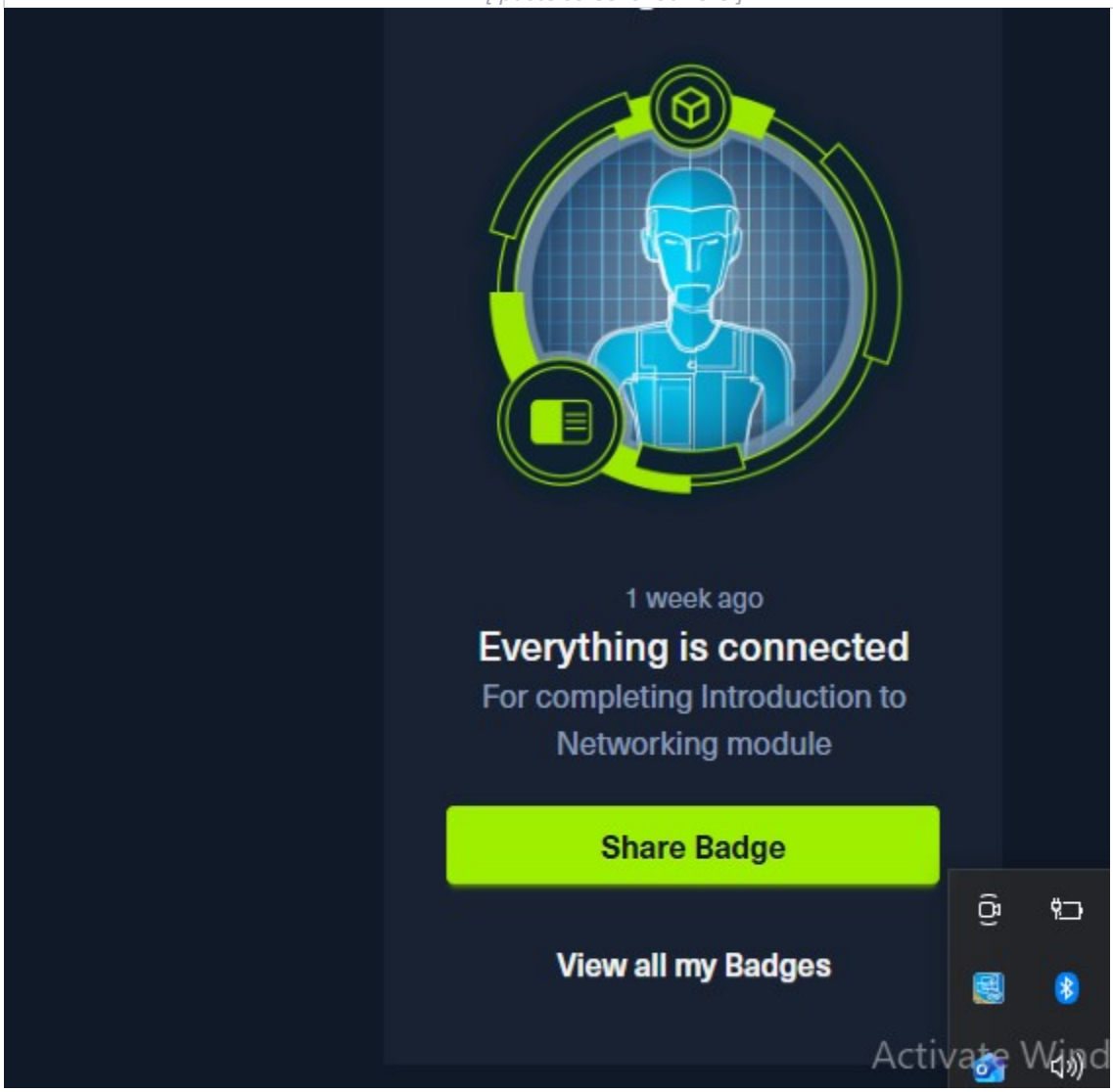
> *Paste the shareable link the platform gives you when you finish, then the completion screenshot.*

Shareable link

https://academy.hackthebox.com/achievement/1738064/34

Completion screenshot

[paste screenshot here]



2. Introduction

This module provided a comprehensive foundation in network architecture, protocols,

and basic mathematical calculation frameworks for subnets. Understanding these

core networking concepts is absolutely vital for a security analyst because every

adversarial action, malicious payload, and defensive telemetry signal depends entirely

on network pathways. Prior to beginning this module, I expected to gain clarity

on how data is segmented and routed, as well as how to systematically calculate

subnet barriers to limit network visibility and isolate potential host exposures

3. The work

3.1 3.1 Subnetting

What this section covered:

This section explored the methodology of dividing a contiguous block of IPv4 addresses into smaller, isolated logical network spaces. It covered the mechanics of separating the network part from the host part using custom subnet masks, determining network boundaries, and calculating host ranges manually using powers of two and modulo operati

Question 1

Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27

Answer

255.255.255.224

How you got there

1. The CIDR notation specifies that the first 27 bits are turned "on" (1s) to represent the network portion.

2. Writing this out in binary across 4 octets yields: 11111111.11111111.11111111.11100000.

3. Converting each binary octet into decimal notation gives:

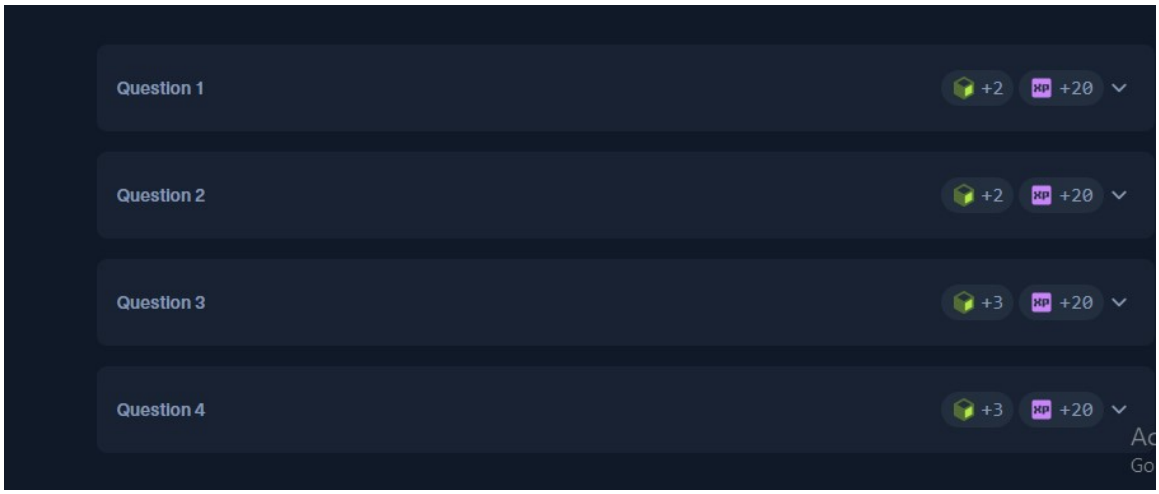
*** 1st, 2nd, and 3rd Octet = 255**

*** 4th Octet = 128 + 64 + 32 = 224**

4. Combining the segments yields the complete decimal subnet mask: 255.255.255.224.

Evidence

The screenshot shows the Hack The Box Academy interface for Section 10 with a green checkmark next to Question 1.



Question 2

Submit the broadcast address of the following CIDR: 10.200.20.0/27

Answer

10.200.20.31

How you got there

1. A /27 subnet mask uses 27 bits for the network, leaving 5 bits remaining for the host portion (32 - 27 = 5).

2. The total block size size is calculated as 2^5 = 32 addresses.

3. Since the network address starts at 10.200.20.0, the total address span for this block runs from .0 to .31.

4. To find the broadcast address, we flip all 5 host bits to 1s, which gives the very last address in the pool: 10.200.20.31.

Evidence

The screenshot captures the completed field input for Question 2 with a verified submission status indicator.

It proves that the boundary analysis correctly isolates the terminal broadcast address for the block.

Question 3

Split the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer.

Answer

10.200.20.16

How you got there

1. To split an existing network space into 4 smaller subnets, we apply the formula: 2^n >= 4. This means we must borrow an additional 2 bits (2^2 = 4).

2. The new classless subnet mask increases from the baseline /27 by 2 bits, becoming /29.

3. With a /29 mask, the remaining host bits drop to 3 (32 - 29 = 3), resulting in a precise increment block size of 2^3 = 8 addresses per subnet.

4. Starting from the original base address (.0), we step up sequentially by 8 to map our subnets:

* *Subnet 1 Network Address: 10.200.20.0*

* *Subnet 2 Network Address: 10.200.20.8*

* *Subnet 3 Network Address: 10.200.20.16*

* *Subnet 4 Network Address: 10.200.20.24*

Question 4

Split the network 10.200.20.0/27 into 4 subnets and submit the broadcast address of the 2nd subnet as the answer.

Answer

10.200.20.15

How you got there

1. Using our calculations from Question 3, each of the four newly created /29 subnets has a block step size of 8 addresses.

2. Subnet 2 begins exactly at the Network ID address 10.200.20.8.

3. Subnet 3 begins directly after it at the Network ID address 10.200.20.16.

4. Since the broadcast address of a subnet is always exactly one increment position below the next network's ID address, we simply subtract 1 from Subnet 3's starting point: 16 - 1 = 15. This yields a final broadcast parameter of 10.200.20.15.

4. Challenges and how you resolved them

The primary challenge faced during this module was tracking custom binary bit

borrowing boundaries without making calculation errors in the final octet.

Initially, I mixed up host capacities by forgetting to subtract the Network ID

and Broadcast addresses. I resolved this by drafting out the block sizes manually

on paper and cross-verifying the boundaries using structural mathematical checks

to make sure no IP spaces overlapped.

5. Conclusion

Completing this module has significantly advanced my engineering perspective

from blindly running scanning utilities to understanding how packet data travels

across network interfaces. I can now confidently analyze a classless prefix, map

active listening ports to their underlying services, and calculate functional

subnetworks. This structural clarity will immediately help me interpret raw traffic

captures, configure firewall boundaries, and trace data movements as a security